

## UNIT 2 - DEVELOPMENT OF CHEMISTRY



### DEVELOPMENT OF CHEMISTRY—A HISTORICAL PERSPECTIVE

Magic is something which cannot be explained by scientific reason. Earlier people thought that alchemists were wizards who could call spirits. The alchemists wanted to fool the people by using some strange words like "Hocus pocus" to convert a base metal into gold.

The word "**Alchemy**" originated from Ancient Greek word "Khumeria". Which is an ancient branch of natural philosophy.

The people who are practising alchemy were considered to be early chemists and were known as "**Alchemists**". Alchemists were attempted to purify, mature and perfect certain materials.

In India, Alchemy was also practised for medicines which is now known as **Ayurvedic medicines**.

In the 16th century the alchemist **Paracelsus** promoted the experimental aspects over the spiritual ones. In 17th century **Glauber** discovered **sodium sulphate**, which was named after him and called Glauber's salt. He also prepared **hydrochloric acid**. Chinese alchemists discovered gun powder in 9th century by mixing powdered **charcoal**, **sulphur** and **nitre** in certain fixed proportion. Gun powder is used in fire works. In 8th century alchemist **Jabir Ibn** extracted some compounds from herbs which were used as **general anaesthesia** during operations on humans.



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### FOOD AND CHEMISTRY

Food science deals with the production, processing, preparation and utilisation of food. Food chemists work with plants that have been harvested for food. They also work with animals which are useful to humans. Chemistry helps in the production of safe and healthy food in the following ways.

- **Animal Chemists**  
Animal chemists conduct research on animals which are useful to humans.
- **Flavour Chemists**  
Flavour chemists use knowledge of the chemistry of food ingredients and **create new and improved flavours**.
- **Food Chemists**  
Food chemists help in processing, packaging, preserving, storing and distributing foods and drinks to make them safe.

#### DID YOU KNOW?

Base metal is a cheaper metal than gold or silver. For example, zinc, iron, lead, tin, etc., are base metals.

#### DID YOU KNOW?

Glauber heated common salt with oil of vitriol (sulphuric acid) in a glass flask and obtained sodium sulphate (a salt known by his name) and hydrochloric acid gas. This gas on dissolving in water formed hydrochloric acid.

#### DID YOU KNOW?

When alcohol is treated with organic acids such as acetic acid, propanoic acid, it forms a compound which have a fruity smell. These compounds are commonly called esters.



## SPOTLIGHT

The iron pillar near Chhatrapati Shivaji Maharaj was made around 1215 AD. It is a marvel of Indian metallurgy achieved by the Indian metal workers. It is 8 m tall and weighs 6 tonnes. This pillar has not rusted in the last 16 centuries.



It is likely because of the formation of a thin film of magnetic oxide of iron ( $\text{Fe}_3\text{O}_4$ ) on its surface, as a result of the finishing treatment given to the pillar, by painting it with a mixture of different salts, then heating and quenching in water.



Fig. 1.3 : Rayon fabric



Fig. 1.4 : Nylon articles

### ■ Nutritional Chemists

Nutritional Chemists perform research on the physical and chemical properties of nutrients.

### ■ Soil and Plant Chemists

Soil and Plant Chemists examine the **composition of soil** and its effects on plant growth and develop methods to **conserve and manage the fertility of soil**. They help farmers by providing them with agro-chemicals like **fertilisers, pesticides, insecticides and fungicides**.

## COSMETICS AND CHEMISTRY



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Cosmetics are an excellent example of how discoveries in chemistry are part of our day to day lives. All kinds of powders, creams, lipsticks, cleansing lotions, deodorants, nail polishes, soaps, detergents, etc., are cosmetics. These are the result of research done by the chemists.

**Cosmetics are chemical compounds which are used to improve our appearance.** It is not a recent phenomenon. It is said that the Egyptian queen bathed in milk, as a way to keep her skin beautiful and soft.

The different cosmetic products have a strong component of scientific innovation, developed in modern research laboratories. For example, **Talcum powder is made up of magnesium carbonate and zinc oxide.**



TWENTYFIVE

## CLOTHING AND CHEMISTRY

Polymer chemistry has contributed to clothing technology by providing man-made or synthetic fibres to replace natural polymers such as silk, cotton, wool, etc.

### Types of human-made (synthetic) fibre

- (i) **Rayon or Artificial silk** : The **raw material** for the production of rayon is **pure cotton or wood cellulose**. Rayon is sometimes called **regenerated fibre**, because it is produced by modifying the natural fibre. It is not a truly artificial fibre, because its raw material is natural fibre. However, for all practical purposes it is included into human-made fibre or synthetic fibre. The filament of rayon can be woven like any other natural fibres. They have a **feel and texture of natural silk**.
- (ii) **Nylon** : It is a superior type of **synthetic fibre**, which is truly artificial. The nylon fibres are elastic, strong and water resistant. It is used for making fabric for **clothing and parachutes**. It is used for making stockings, ropes for rock climbing and **fishing nets**. It is used for making common household articles such



as tooth brushes, combs, zip-fasteners, hooks, etc. Tyre cords, small parts of machines, such as washers, pulleys, etc., are made from nylon fibres.

- (iii) **Polyesters** : The polyester can be drawn into very fine filaments so as to form artificial fibre. There are a number of varieties of esters. **Terylene** filament is one of them. This filament is similar, but superior in properties than nylon. It is largely used for making fabrics for **shirts, trousers, sarees** and other **dress materials**. Terylene is used for making **sails for sailing boats**, fire hoses and conveyor belts.



Fig. 1.5 : Polyester dresses

### Advantages of Synthetic Fibres

1. They can be easily washed.
2. They get dried quickly.
3. They do not get creased.
4. They are cheaper.

## CHEMICALS AS MEDICINE



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Man has been dependent on medicinal plants right from the dawn of civilisation. He used leaves, roots, seeds or fruits of medicinal plant to cure himself from common diseases. However, these days chemists have developed methods to identify the active molecules in a medicinal plant. Thus, by extracting these molecules from plants, better medicines are prepared. Chemists have also developed methods to prepare such molecules artificially, thereby reducing dependence on medicinal plants. Even new molecules with better properties have been made in laboratory, which has helped to cure a number of incurable diseases. Discovery of **penicillin, tetramycine, sulphadiazine**, etc., are the examples of new medicines developed by the man. **Quinine** is an example of medicine extracted from the bark of a cinchona plant.

Chemistry plays a huge role in the development of synthetic drugs such as **antibiotics, antimalarials** and **analgesics** [pain killers]. New research is being done on possible ways so that chemistry can be used to target cancer cells without killing the healthy cells around the cancer growth.

## CHEMICALS IN INDUSTRIES



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Industries are set up to produce a large number of consumer products. Chemistry has helped in the growth of different industries. Chemicals used in industry to process the raw materials so as to improve their properties are called industrial chemicals. They are used in removing stains.

**Stain removal** : Stain removal is the process of removing a spot left



Fig. 1.6 : Some common medicines



by one substance on a fabric. A solvent or detergent is generally used to remove the stain. On ironing the fabric having stain, it gets chemically bonded to fabric. So ironing of the stain should be avoided.

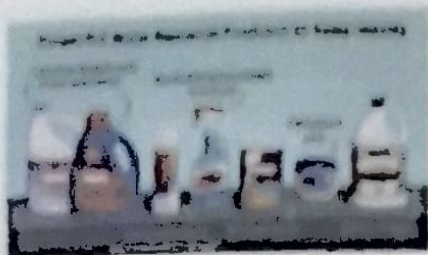


Fig. 1.7 : Some common stain remover

- (i) Household bleach contains hydrogen peroxide which can be used to remove common stains.
- (ii) Sodium hydroxide and sodium hypochlorite are commonly used for removing stains and dye colours.
- (iii) Acetone is good for removing some glues, nail polish, ink stain, rubber and grease. Nail polish remover also contains acetone.
- (iv) Detergents are used for washing clothes and removing stains, oil spots, etc.
- (v) Lemon juice contains citric acid and acts as bleaching agent. It can remove stains. Its action can be accelerated by exposing the stain to sunlight. Boric acid, vinegar (acetic acid) are used to remove stains caused by rust.
- (vi) Sodium hydroxide is also commonly used in drain cleaning. It allows grease and other oils to dissolve in water. Potassium hydroxide is another alkali which is also used in this regard.
- (vii) Glycerine can be used to soften "set" stains, especially on wool and non-water washable fabrics.
- (viii) Boiling water can be used to take out fruit juice stains.

## DARK SIDE OF CHEMISTRY

There is no doubt that knowledge of chemistry helps us to produce certain chemicals which are very useful in maintaining health services, in the production of food or for national defence. However, the misuse of such chemical bring about massive destruction. Some examples are as follows:

1. **Chemical weapons:** Chemistry has helped to produce deadly explosives such as tri-nitro-toluene (T.N.T.), tri-nitro-glycerine (T.N.G) and Rapid denonating explosive (RDX), which are used as main ammunition for army. However, the same explosives in the hand of terrorists bring about lot of misery.
2. **Pollutants:** Chemicals such as BHC (benzene hexachloride) and DDT (dichloro-diphenyl trichloro ethane) are widely used in controlling mosquitoes. However, these chemical pollute air, water and soil. They enter in food chain and cause serious diseases in human beings.
3. **Addictive drugs:** Drugs like, cocaine, morphine, heroin, etc., are useful in removing anxiety or pain under very strict medical control. However, same drugs without medical control become addictive and can kill a person, if someone takes overdose by mistake.



Fig. 1.8 : Chemical weapons



## NOTABLE PERSONS AND THEIR CONTRIBUTIONS



Henry Cavendish

### 1. Henry Cavendish (1731-1810)

**Henry Cavendish** was born in Nice, France, on October 10, 1731. He discovered hydrogen gas and the composition of water contains two parts of hydrogen and one part of oxygen by volume. He prepared "inflammable air" (hydrogen) by the action of dilute acids on metals. He also investigated the products of fermentation, a chemical reaction that splits complex organic compounds into simple substances. He discovered the reversible reaction between calcium carbonate and carbon dioxide to form calcium bicarbonate, which causes temporary hardness of water. He also found out how to soften such water by adding lime (calcium hydroxide).

### 2. Antoine Lavoisier (1743-1794)



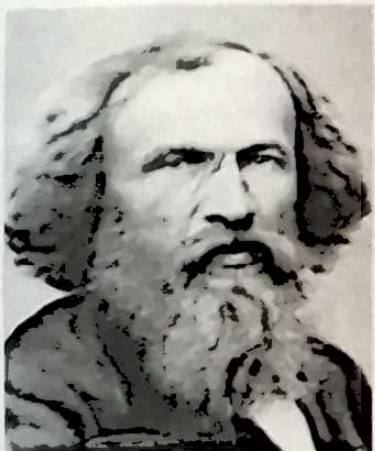
Antoine Lavoisier

**Antoine Lavoisier** was born in Paris (France) on August 26, 1743. Antoine Lavoisier revolutionised chemistry. He named the elements carbon, hydrogen and oxygen. He discovered oxygen's role in combustion and respiration. He also established that water is a compound of hydrogen and oxygen; and he discovered that sulphur is an element.

He gave the Law of Conservation of Mass.

He made a clear difference between elements and compounds. He made a list of chemical elements which included oxygen, nitrogen, hydrogen, sulphur, phosphorus, carbon, antimony, cobalt, copper, gold, iron, manganese, molybdenum, nickel, platinum, silver, tin, tungsten, and zinc.

### 3. Dmitri Mendeleev (1834-1907)



Dmitri Mendeleev

**Dmitri Ivanovich Mendeleev** was born on February 8, 1834 in Verkhnie Aremzyani, in the Russian province of Siberia.

In 1860, Mendeleev attended the first International Chemistry Conference, which took place in Germany. Mendeleev's periodic table was based on atomic weights.

He wrote the name of the 65 known elements on cards—one element on each card. Then he wrote the properties of each element on the card. He found that the properties of the elements get repeated with the increase in atomic weight. He arranged the elements in increasing order of their atomic weight in the form of a table which he called Periodic Table.

He proposed that atomic weight of some elements were wrongly measured. It was later found to be correct. Also he predicted the





Robert Boyle



John Dalton

existence of eight new elements and their properties. Later on these elements were discovered. In his honour an element which was discovered later was named as Mendeleevium.

#### 4. Robert Boyle (1627-1691)

**Famous as** being the first Modern Chemist. He is also known as Father of Modern Chemistry.

He was born in Ireland. In 1654, after moving to Oxford he worked on gases using an air pump made by Robert Hooke. He proved that sound does not travel in a vacuum. In 1662, he discovered Boyle's Law. He worked on flame tests and acid-base indicators in chemistry. He defined elements, compounds and mixtures.

#### 5. John Dalton (1766-1844)

**Famous for** identification and presenting the atomic theory. He was a British chemist and physicist. But he is the best recognised for his work on the Atomic Theory and research on colour blindness. He successfully identified chemical compounds and reactions affected by interaction of atoms with one another.

## Exercise 1.2

Fill in the blank spaces by choosing the correct words from the list given below:

List: Clothing, reason, innovation, Rayon, Glauber's

1. Magic is something which cannot be explained by scientific reason.
2. Sodium sulphate is also known as Glauber's salt.
3. The different cosmetic products have a strong component of scientific innovation.
4. Polymer chemistry has contributed to clothing technology.
5. Rayon is sometimes called regenerated fibre.

Statements given below are incorrect. Write correct statements:

1. Flavour chemists use knowledge of the chemistry of fragrance and create new and improved perfume for mankind.
2. Detergents are used to remove stains caused by rust.
3. Nylon has a feel and texture of natural silk.
4. Rayons are compounds formed, when alcohol is made to react with organic acids.
5. Quinine is an example of medicine made from the animals product.

Match the statements:

- |                                                                                                                      |                       |
|----------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1. Something which cannot be explained by scientific reason. <u>c</u>                                                | (a) John Dalton       |
| 2. A person which has contributed to the study of processing of food. <u>e</u>                                       | (b) Rayon             |
| 3. The fibre known as regenerated fibre. <u>b</u>                                                                    | (c) Magic             |
| 4. A branch of chemistry which has also contributed to clothing technology. <u>d</u>                                 | (d) Polymer chemistry |
| 5. A chemist who is the best recognised for his work on the Atomic Theory and research on colour blindness. <u>a</u> | (e) Food chemist      |



D. Write true or false for the following statements :

1. Food chemists work with plants that have been harvested for food and animals which are useful to humans.
2. Man has been independent on medicinal plants in ancient time.
3. A solvent or detergent is generally used to remove stain.
4. Forensic scientists work for Finance Department for giving proper advice regarding economic policy.
5. Henry Cavendish discovered hydrogen and composition of water.

False  
True  
True  
\_\_\_\_\_  
\_\_\_\_\_

E. Tick [✓] the most appropriate answer.

1. Which of the following chemical compounds is used to improve our appearance?  
[a] Medicines [b] Cosmetics [c] Adhesives [d] All of these
2. Which of the following artificial fibres is known as artificial silk?  
[a] Rayon [b] Nylon [c] Polyester [d] Acrylon
3. The analgesic drug is generally known as  
[a] antibiotic [b] paracetamol [c] pain killer [d] antimalarial
4. Which of the following chemists discovered oxygen's role in combustion and respiration?  
[a] Henry Cavendish [b] Antoine Lavoisier [c] Dmitri Mendeleev [d] Robert Boyle
5. Which of the following chemists recognised for his work in atomic theory and research on colour blindness?  
[a] John Dalton [b] Robert Boyle [c] Henry Cavendish [d] Antoine Lavoisier

F. Answer the following questions:

1. Write short notes on development of chemistry – a historical perspective.
2. How does chemistry help in the production of food?
3. How can you say chemistry plays a vital role in the production of cosmetics?
4. [a] Define polymer chemistry.  
[b] Write at least three synthetic fibres and explain their properties and uses.
5. How does chemistry help in medical applications? Explain properly.
6. [a] Define industrial chemistry.  
[b] What do you mean by stain removal? Write at least three stain removers and their proper applications in removing the stain.

## KEYWORDS

1. **Chemistry** : The branch of science which deals with the study of materials (substances); especially about their composition, methods of preparation, properties and their reactions with other substances.
2. **Food Science** : A science which deals with the production, processing, preparation and utilisation of food.
3. **Cosmetics** : The chemical compounds which are used to improve our appearance and also help us to live in a clean environment.
4. **Polymer Chemistry** : The chemistry which has contributed to clothing technology by providing man-made to replace natural polymers such as silk, cotton, wool, etc. Rayon, nylon and polyester are examples of some man-made fibres.



- 5. Industrial Chemistry** : A branch of chemistry which has contributed to prepare the useful products from raw materials by using some physical and chemical processes.
- 6. Stain Removal** : A process of removing a spot left by some substances on a fabric.

## HIGHLIGHTS IN REVIEW

1. The branch of science which deals with the study of materials, especially about their composition, methods of preparation, properties and their reactions with other substances is called chemistry.
2. From materials we mean, common substances around us such as, air, water, trees, stones, soil, etc.
3. Chemistry plays a very important role in all kinds of human activities.
4. A solvent or detergent is generally used to remove the stain.
5. **Henry Cavendish** discovered hydrogen and the composition of water.
6. **Antoine Lavoisier** revolutionised chemistry. He named the elements carbon, hydrogen and oxygen. He discovered oxygen's role in combustion and respiration and sulphur as an element. He also established that water is a compound of hydrogen and oxygen.
7. **Dmitri Mendeleev** formed a periodic table which was based on atomic weights. He proposed that atomic weight of some elements were wrongly measured. It was latter found to be correct. Also he predicted the existence of eight new elements and their properties.
8. **Robert Boyle** proved that sound does not travel in vacuum. He discovered Boyle's law. He worked on flame test and acid-base indicator.
9. **John Dalton** is the best recognised for his work on the atomic theory and research on colour blindness.

## THEME ASSIGNMENTS

### A. Fill in the blank spaces:

- (i) Chemistry has led us to develop \_\_\_\_\_, which in turn help in conserving forests. [rubber/plastics]
- (ii) Artificial \_\_\_\_\_ have helped farmers to grow more food. [fertilisers/fungicides]
- (iii) Cosmetics are \_\_\_\_\_ compounds which are used to improve our appearance. [physical/chemical]

### B. Write true or false for the following statements.

- (i) Herbicides kills insects. \_\_\_\_\_
- (ii) Terylene is a human-made product. \_\_\_\_\_
- (iii) All kinds of cosmetics are the result of research in chemistry. \_\_\_\_\_

### C. Tick (✓) the most appropriate answer :

- (i) What is the backbone of an Industrialised country?  
 (a) Plastic (b) Fertilisers (c) Cotton (d) Steel
- (ii) LPG is :  
 (a) Liquefied Petrol Gas (b) Liquid Petrol Gas  
 (c) Liquefied Petroleum Gas (d) Liquid Petroleum Gas
- (iii) Which monument was made around 1215 AD?  
 (a) Qutub Minar (b) Taj Mahal (c) Hawa Mahal (d) Amer Fort
- (iv) Rayon is produced from :  
 (a) cotton (b) silk (c) paper (d) wool
- (v) Which chemical is used to remove glues?  
 (a) NaOH (b) Glycerine (c)  $\text{Fe}_2\text{O}_3$  (d) Acetone



D. Statements given below are incorrect. Write the correct statements.

- (i) Pesticides fight variety of diseases caused by bacteria.
- (ii) Nylon is an improved natural product.
- (iii) Rayon is water proof as well as air proof.

#### E. Assertion-Reason Type Questions

In the questions given below, there are two statements marked as Assertion [A] and Reason [R]. Read the statements and choose the correct option.

- (a) If both Assertion [A] and Reason [R] are correct and Reason [R] is the correct explanation of Assertion [A].
- (b) If both Assertion [A] and Reason [R] are correct but Reason [R] is Not the correct explanation of Assertion [A].
- (c) If Assertion [A] is true but Reason [R] is false.
- (d) If Assertion [A] is false but Reason [R] is true.

(i) **Assertion [A]** : Moist air corrodes iron.

**Reason [R]** : Rusting happens in the presence of moisture.

(ii) **Assertion [A]** : Polyesters are hard materials.

**Reason [R]** : Polyesters are used to form artificial fibre.

(iii) **Assertion [A]** : Tyre cord is made up of nylon.

**Reason [R]** : Combs are made from nylon.

#### F. Case Study Based Questions

Read the given information carefully and answer the questions that follow:

There are many industries which use chemicals to improve the properties of the product. These chemicals are used with raw materials. Many chemicals are used for removing stains, to dye clothes etc. Chemistry has helped in the growth of different industries.

- (i) Which acid is used to remove stain caused by rust?  
(a) Acetic acid                      (b) Oxalic acid                      (c) Citric acid                      (d) Both [a] and [c]
- (ii) Nail Polish remover contains  
(a) acetone                      (b) NaOH                      (c) potassium hydroxide                      (d) none of these
- (iii) Dyes, colours are removed with:  
(a) hydrogen peroxide                      (b) sodium hydrosulphite                      (c) sodium hypochlorite                      (d) both [b] and [c]

#### G. Answer the following questions:

1. (a) Name two medicines which fight against bacterial diseases.  
(b) Name two human-made products used for making dresses.
2. What do you mean by cosmetics? How does chemistry become helpful in this regard?
3. (a) Write short note on polymer chemistry.  
(b) Write three synthetic fibres and explain their properties and uses.
4. List at least three notable persons and their contributions to chemistry. Explain briefly.

#### H. Problem-Solving Skill Based Question

There are many medicines which are not good for health and have many side effects but doctors still refer those medicines. This type of activity is seen in hospitals also. What steps you will take to stop this activity?

#### I. Art-Integration Based Question

Draw the different articles used in chemistry laboratory on chart paper.

#### J. Inquiry Based Questions

- (i) Inquire about the chemicals used in manufacturing of soaps and detergents.
- (ii) How is nail polish harmful for nails?